Andrew J. Peng

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EDUCATION

Bachelor of Science in Computer Engineering Aug. 2024 – Aug. 2025
University of Maryland College Park, MD

- **GPA**: 3.96 | A. James Clark School of Engineering Academic Honors

EXPERIENCE

Machine Learning Research Assistant May 2026 – Aug. 2026
Shanghai Jiao Tong University Shanghai, China

- Conducted **few-shot object detection** research for satellite imagery on a **4,481-image dataset** spanning **25 classes** of ships, military aircraft, and launch vehicles, supervised by **Prof. Hongtian Chen**, collaborating with graduate researchers.
- Benchmarked **RF-DETR Small** using **COCO-style** whole-image evaluation, **worst-image ranking**, and **SAHI/tiled inference** comparisons to identify the most reliable detection strategy for **small, few-shot** classes.
- Diagnosed model accuracy errors through per-class and confusion-matrix analysis, uncovering **dataset labeling inconsistencies**; used **DarkLabel** to manually review and correct mislabeled annotations.

Game Developer Dec. 2024 – Present
Contractor Belmont, CA

- Utilized **Luau** to design and deliver **30+ modular game systems**, including data management, UI/UX, gameplay mechanics, and physics simulations for client commissions; **contributing to 15,000+ cumulative visits** on **Roblox**.
- Contracted for Tiny Lab Studio, developing **data management systems** and **core gameplay mechanics**, including **exploit security** such as rate limiting and server-authoritative action validation; supporting a **4,000+ member community**.

Vehicle Dynamics & Ergonomics Engineer Aug. 2024 – May 2026
Terps Racing College Park, MD

- Designed **ABS housings** using **SolidWorks** and integrated **brake temperature sensors** and **push rod strain gauges** for the 2026 FSAE EV; conducted FEA on sensor mounts to validate structural integrity against vibrational racing loads.
- Modeled a **clutch actuation** assembly using **SolidWorks** for the 2025 FSAE IC vehicle; conducted FEA on the Onyx carbon lever and 6061 aluminum mount to validate structural integrity under **150 N peak input** and vibrational loads; contributed to vehicle manufacturing via **water jet machining** and **3D printing**.

PROJECTS

ATLAS: Automated Training, Labeling & Annotation System Apr. 2026 – July 2026

- Built a **computer vision tool** that streamlines constructing and deploying a custom object detector, with a **CLI-driven** workflow (dataset version, model size, hyperparameters) powered by **YOLOv8** and **PyTorch**.
- Developed a **semi-automated annotation tool** that uses a pretrained model to suggest bounding boxes, reviewed and corrected through a custom **OpenCV** UI, and exported as **YOLO-format** labels.
- Configured **transfer learning** fine-tuning with **mixed-precision training**, data augmentation, early stopping, and checkpoint resuming, with automatic **device selection** across CUDA/MPS/CPU.

ProfileLite Dec. 2025 – Jan. 2026

- Authored a **type-safe** Roblox DataStore wrapper providing **session-locking**, **auto-saving**, and **cross-server conflict resolution**, using a **ProfileService**-style API; adopted by other developers.
- Implemented an automatic **mock-store** fallback for API-disabled Studio development and paginated **version-history** querying, utilizing a **custom**, **yield-safe** signal system for reliability diagnostics.

SKILLS

Tools: Git, DarkLabel, SolidWorks, Fusion 360, Arduino, Raspberry Pi
Languages: Python, Luau, Java, MATLAB, HTML/CSS
Frameworks: PyTorch, OpenCV, YOLOv8, RF-DETR